

Problem Set 3: Two-Period New Keynesian Model

Macroeconomics II (Spring 2026)

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Problem 1 (Questions 1–20): IS / Euler block

- ① Timeline in two-period NK. Which expectation operator is nontrivial? _____
- (A) $\mathbb{E}_1[\cdot]$ about period 2 variables
 - (B) $\mathbb{E}_0[\cdot]$ about period 1 variables
 - (C) $\mathbb{E}_{-1}[\cdot]$ about period 0 variables
 - (D) No expectations appear in a two-period model
- ② Fisher relation (ex-ante real rate). Which is correct? _____
- (A) $r_0 = i_0 + \mathbb{E}_0\pi_1$
 - (B) $r_0 = \mathbb{E}_0\pi_1 - i_0$
 - (C) $r_0 = i_0 - \mathbb{E}_0\pi_1$
 - (D) $r_0 = i_0 - \pi_0$
- ③ NK IS (Euler) equation: pick the standard form. _____
- (A) $x_0 = \mathbb{E}_0x_1 - \frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$
 - (B) $x_0 = \mathbb{E}_0x_1 + \frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$
 - (C) $x_0 = \frac{1}{\sigma}(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$
 - (D) $x_0 = \mathbb{E}_0x_1 - \sigma(i_0 - \mathbb{E}_0\pi_1 - r_0^n)$
- ④ Meaning of σ in NK IS. _____

- (A) Policy response to inflation
- (B) Price rigidity parameter
- (C) Inverse intertemporal elasticity of substitution
- (D) Slope of Phillips curve

⑤ If $x_1 = 0$ for sure, NK IS implies: _____

- (A) $x_0 = \mathbb{E}_0 x_1$
- (B) $x_0 = -\frac{1}{\sigma}(i_0 - \mathbb{E}_0 \pi_1 - r_0^n)$
- (C) $x_0 = +\frac{1}{\sigma}(i_0 - \mathbb{E}_0 \pi_1 - r_0^n)$
- (D) $x_0 = 0$ always

⑥ Numerical IS. $\sigma = 2$, $r_0^n = 1\%$, $i_0 = 3\%$, $\mathbb{E}_0 \pi_1 = 1\%$, $x_1 = 0$. Then $x_0 =$ _____

- (A) $+0.5\%$
- (B) 0
- (C) -0.5%
- (D) -1%

⑦ Natural real rate r_0^n is: _____

- (A) The nominal policy rate
- (B) Inflation target
- (C) Real rate consistent with $x_0 = 0$ under sticky prices
- (D) Real rate consistent with $x_0 = 0$ under flexible prices

⑧ Holding i_0 fixed, higher $\mathbb{E}_0 \pi_1$ implies: _____

- (A) Higher r_0 and lower x_0
- (B) Lower r_0 and higher x_0
- (C) Lower r_0 and lower x_0
- (D) No effect on x_0

9 ZLB constraint is: _____

- (A) $r_0 \geq 0$
- (B) $\pi_0 \geq 0$
- (C) $i_0 \geq 0$
- (D) $x_0 \geq 0$

10 At ZLB, if expected inflation falls, then: _____

- (A) Real rate rises and demand falls
- (B) Real rate falls and demand rises
- (C) Nominal rate rises automatically
- (D) NKPC shifts down only

11 NK Phillips curve (NKPC) form is: _____

- (A) $\pi_0 = \pi_{-1} + \kappa x_0 + u_0$
- (B) $\pi_0 = \beta \mathbb{E}_0 \pi_1 + \kappa x_0 + u_0$
- (C) $\pi_0 = \beta \pi_1 - \kappa x_0 + u_0$
- (D) $\pi_0 = \kappa x_0 + u_0$

12 Interpretation of β in NKPC: _____

- (A) Discount factor weight on expected inflation
- (B) Policy response to inflation
- (C) IS slope parameter
- (D) Cost-push shock size

13 Interpretation of u_0 in NKPC: _____

- (A) Cost-push/markup shock
- (B) Demand shock
- (C) Natural rate
- (D) Nominal rate

⑭ If $\mathbb{E}_0\pi_1 = 0$, NKPC becomes: _____

- (A) $\pi_0 = \kappa x_0 + u_0$
- (B) $\pi_0 = \beta + \kappa x_0 + u_0$
- (C) $\pi_0 = \beta\pi_0 + \kappa x_0 + u_0$
- (D) $\pi_0 = u_0$

⑮ Taylor rule (simple) is: _____

- (A) $i_0 = \bar{i} + \phi_\pi\pi_0 + \phi_x x_0$
- (B) $i_0 = r_0^n$
- (C) $i_0 = \pi_0 - \phi_\pi x_0$
- (D) $i_0 = \mathbb{E}_0\pi_1$

⑯ Taylor principle typically requires: _____

- (A) $\phi_\pi < 1$
- (B) $\phi_\pi = 1$
- (C) $\phi_\pi > 1$
- (D) $\phi_\pi < 0$

⑰ Determinacy intuition: when inflation rises, real rate should: _____

- (A) Rise (tighten)
- (B) Fall (loosen)
- (C) Stay constant
- (D) Be undefined

⑱ Demand shock (IS) tends to move: _____

- (A) $x \uparrow, \pi \uparrow$
- (B) $x \downarrow, \pi \uparrow$
- (C) $x \uparrow, \pi \downarrow$
- (D) $x \downarrow, \pi \downarrow$

⑲ Cost-push shock ($u_0 > 0$) tends to move: _____

- (A) $x \uparrow, \pi \uparrow$
- (B) $x \downarrow, \pi \uparrow$
- (C) $x \uparrow, \pi \downarrow$
- (D) $x \downarrow, \pi \downarrow$

⑳ Larger κ means (holding x fixed): _____

- (A) Inflation responds less
 - (B) Inflation responds more
 - (C) No change
 - (D) Inflation becomes backward-looking
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Problem 2 (Questions 21–40): NKPC + Taylor rule

㉑ Numerical NKPC: $\beta = 1$, $\mathbb{E}_0 \pi_1 = 1\%$, $\kappa = 0.5$, $x_0 = 2\%$, $u_0 = 0$. Then $\pi_0 =$ _____

- (A) 1%
- (B) 2%
- (C) 2%
- (D) 0.5%

(22) Numerical Taylor: $ari = 1\%$, $\phi_\pi = 1.5$, $\phi_x = 0$, $\pi_0 = 2\%$. Then $i_0 =$ _____

- (A) 2%
- (B) 3%
- (C) 4%
- (D) 1%

(23) Compute real rate: $i_0 = 4\%$, $\mathbb{E}_0\pi_1 = 1\%$. Then $r_0 =$ _____

- (A) 3%
- (B) 5%
- (C) -3%
- (D) 1%

(24) If CB raises i_0 holding expectations fixed, IS implies x_0 : _____

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Undefined

(25) If terminal conditions are $x_1 = 0$ and $\pi_1 = 0$, then: _____

- (A) $\mathbb{E}_0x_1 = \mathbb{E}_0\pi_1 = 0$
- (B) Expectations become larger
- (C) Expectations are indeterminate
- (D) Expectations equal current values

(26) If ϕ_x increases, policy responds more to: _____

- (A) Inflation only
- (B) Output gap (stabilization)
- (C) Expected inflation only
- (D) Natural rate only

②7 If $\phi_\pi = 0$ (peg), higher inflation tends to: _____

- (A) Raise real rate
- (B) Lower real rate
- (C) Leave real rate unchanged
- (D) Stabilize inflation

②8 With $\phi_\pi > 1$, inflation rise tends to: _____

- (A) Reduce real rate
- (B) Increase real rate
- (C) Not change real rate
- (D) Set $x_0 = 0$ always

②9 If expected inflation rises but rule reacts to current π_0 only, i_0 changes: _____

- (A) Immediately due to expectations
- (B) Only if π_0 changes
- (C) One-for-one always
- (D) Never

③0 Which blocks are forward-looking? _____

- (A) IS only
- (B) NKPC only
- (C) Both IS and NKPC
- (D) Neither

- 31) If $u_0 > 0$ and CB wants lower π_0 , it chooses: _____
- (A) $x_0 > 0$
 - (B) $x_0 < 0$
 - (C) $x_0 = 0$
 - (D) Any x_0
- 32) If β rises toward 1, NKPC becomes: _____
- (A) Less forward-looking
 - (B) More forward-looking
 - (C) Backward-looking
 - (D) Unrelated to expectations
- 33) If $\mathbb{E}_0\pi_1$ rises holding x_0 fixed, π_0 rises by: _____
- (A) 0
 - (B) β times the change
 - (C) $(1 - \beta)$ times the change
 - (D) κ times the change
- 34) If $\kappa = 0$, then inflation is: _____
- (A) Independent of x_0
 - (B) More sensitive to x_0
 - (C) Always equal to target
 - (D) Always zero
- 35) If σ is very large, IS response to real-rate gap is: _____

- (A) Strong
- (B) Weak
- (C) Zero always
- (D) Opposite sign

③⑥ If r_0^n increases holding i_0 and expectations fixed, then x_0 : _____

- (A) Falls
- (B) Rises
- (C) Unchanged
- (D) Depends only on NKPC

③⑦ If CB sets $i_0 = r_0^n + \mathbb{E}_0\pi_1$ and $x_1 = 0$, then $x_0 =$ _____

- (A) > 0
- (B) $= 0$
- (C) < 0
- (D) Undefined

③⑧ If $i_0 = 0$ and $\mathbb{E}_0\pi_1 = -2\%$, then $r_0 =$ _____

- (A) -2%
- (B) 0%
- (C) 2%
- (D) -4%

③⑨ A higher ϕ_π tends to make equilibrium inflation (given shocks): _____

- (A) More sensitive
- (B) Less sensitive
- (C) Unchanged
- (D) Always zero

④0 Anchored expectations roughly means: _____

- (A) $\mathbb{E}_0 \pi_1$ stays near target
- (B) π_0 always zero
- (C) x_0 always zero
- (D) i_0 never changes